Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The area bounded by the curve $y = \cos^{-1}\sin x + \frac{\pi}{2} - \cos^{-1}\cos x$ and the x-axis, where $\frac{\pi}{2} \leq x \leq \pi$, is equal to

(A) π^2

(B) $\frac{\pi^2}{2}$

(C) $\frac{\pi^2}{8}$

(D) $\frac{\pi^2}{4}$

Q.2

The value of $\sin \cot^{-1} \tan \cos^{-1} x$, is

(A) x

(B) $\frac{1}{x}$

(C) 1

(D) 0

Q.3

The domain of the function
$f(x) = \sin^{-1}\left\{\log_2\left(\frac{1}{2}x^2\right)\right\}$ is

(A) $[-2,-1) \cup [1,2]$

(B) $(-2,-1] \cup [1,2]$

(C) $[-2,-1] \cup [1,2]$

(D) $(-2,-1) \cup (1,2)$

Q.4

Let $a, b, c > 0$ and $x = \tan^{-1}\sqrt{\frac{a}{bc}(a+b+c)}$
$y = \tan^{-1}\sqrt{\frac{b}{ca}(a+b+c)}$ and
$z = \tan^{-1}\sqrt{\frac{c}{ab}(a+b+c)}$, then

(A) $\Sigma \tan x \tan y = 1$

(B) $\Sigma \cot x \cot y = 1$

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(C) $\Sigma x = \frac{\pi}{2}$

(D) None of these

Q.5

What is $\sin\left[\cot^{-1}\left\{\cos\left(\tan^{-1} x\right)\right]\right.$ where $x > 0$, equal to?

(A) $\sqrt{\frac{(x^2+1)}{(x^2+2)}}$

(B) $\sqrt{\frac{(x^2+2)}{(x^2+1)}}$

(C) $\frac{\left(x^2+1\right)}{(x^2+2)}$

(D) $\frac{\left(x^2+2\right)}{(x^2+1)}$

Q.6

The solution set of the equation $\sin^{-1} x = 2\tan^{-1} x$ is

(A) $\{1, 2\}$

(B) $\{-1, 2\}$

(C) $\{-1, 1, 0\}$

(D) $\{1, 1/2, 0\}$

Q.7

If $\sum_{i=1}^{2n} \cos^{-1} x_i = 0$ then $\sum_{i=1}^{2n} x_i$ is

(A) n

(B) $2n$

(C) $\frac{n(n+1)}{2}$

(D) None of these

Q.8

The range of $y = \left(\cot^{-1} x\right)\left(\cot^{-1}(-x)\right)$ is

(A) $\left(0, \frac{\pi^2}{4}\right]$

(B) $(0, \pi)$

(C) $(0, 2\pi]$

(D) $(0, 1]$

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Q.9

$\sum_{r=1}^{n} \sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r(r+1)}}\right)$ is equal to

(A) $\tan^{-1}(\sqrt{n}) - \frac{\pi}{4}$

(B) $\tan^{-1}(\sqrt{n+1}) - \frac{\pi}{4}$

(C) $\tan^{-1}(\sqrt{n})$

(D) $\tan^{-1}(\sqrt{n+1})$

Q.10

What is the value of x that satisfies the equation $\cos^{-1} x = 2\sin^{-1} x$?

(A) $\frac{1}{2}$

(B) -1

(C) 1

(D) $-\frac{1}{2}$

Q.11

If $\left[\sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1} x\right] = 1$, where $[.]$ denotes the greatest integer function, then x belongs to the interval

(A) $[\tan\sin\cos 1, \tan\sin\cos\sin 1]$

(B) $(\tan\sin\cos 1, \tan\sin\cos\sin 1)$

(C) $[-1, 1]$

(D) $[\sin\cos\tan 1, \sin\cos\sin\tan 1]$

Q.12

If $\sin^{-1} x = \tan^{-1} y$, what is the value of $\frac{1}{x^2} - \frac{1}{y^2}$?

(A) 1

(B) -1

(C) 0

(D) 2

Q.13

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If $ax + b\left(\sec\left(\tan^{-1} x\right)\right) = c$ and $ay + b\left(\sec.\left(\tan^{-1} y\right)\right) = c$, then $\frac{x+y}{1-xy} =$

(A) $\frac{ac}{a^2+c^2}$

(B) $\frac{2ac}{a-c}$

(C) $\frac{2ac}{a^2-c^2}$

(D) $\frac{a+c}{1-ac}$

Q.14

$\tan\left\{\frac{1}{2}\sin^{-1}\frac{2x}{1+x^2} + \frac{1}{2}\cos^{-1}\frac{1-y^2}{1+y^2}\right\} =$

(A) $\frac{x-y}{1+xy}$

(B) $\frac{x+y}{1-xy}$

(C) $\frac{x-y}{x+y}$

(D) $\frac{1-xy}{1+xy}$

Q.15

The $+$ ve integral solution of $\tan^{-1} x + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \sin^{-1}\frac{3}{\sqrt{10}}$ is

(A) $x = 1, y = 2; x = 2, y = 7$

(B) $x = 1, y = 3; x = 2, y = 4$

(C) $x = 0, y = 0; x = 3, y = 4$

(D) None of these

Answers & Solutions

Q.1 Answer:

$\frac{\pi^2}{4}$

Solution:

$y = \frac{\pi}{2} - \sin^{-1}\sin x + \frac{\pi}{2} - \cos^{-1}\cos x$

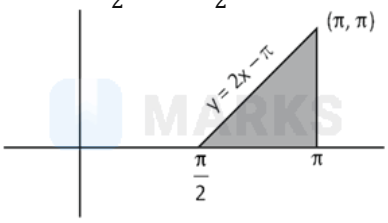
For $x \in \frac{\pi}{2}, \pi$

$y = \frac{\pi}{2} - \pi - x + \frac{\pi}{2} - x$

$y = x - \frac{\pi}{2} + \frac{\pi}{2} - x$

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$y = x - \frac{\pi}{2} + x - \frac{\pi}{2} = 2x - \pi$



Area $= \frac{1}{2} \times \frac{\pi}{2} \times \pi$

$= \frac{\pi^2}{4}$

Q.2 Answer:

x

Solution:

$$\text{Let } \cos^{-1} x = \theta \Rightarrow x = \cos\theta \text{ or } \sec\theta = \frac{1}{x}$$
$$\Rightarrow \tan\theta = \sqrt{\sec^2\theta - 1} = \sqrt{\frac{1}{x^2} - 1} = \frac{1}{|x|}\sqrt{1 - x^2}$$
$$\text{Now, } \sin\cot^{-1}\tan\theta = \sin\cot^{-1}\left(\frac{1}{|x|}\sqrt{1 - x^2}\right)$$
$$\text{Again, putting } x = \sin\theta \text{ , we get}$$
$$\sin\cot^{-1}\left(\frac{1}{|x|}\sqrt{1 - x^2}\right) = \sin\cot^{-1}\left(\frac{\sqrt{1 - \sin^2\theta}}{\sin\theta}\right)$$
$$= \sin\cot^{-1}|\cot\theta| = \sin\theta = x$$

Q.3 Answer:

$[-2, -1] \cup [1, 2]$

Solution:

For $f(x)$ **to be defined, we must have** $-1 \le \log_2\left(\frac{1}{2}x^2\right) \le 1 \Rightarrow 2^{-1} \le \frac{1}{2}x^2 \le 2^1$

$\Rightarrow 1 \le x^2 \le 4$

Now, $1 \le x^2 \Rightarrow x^2 - 1 \ge 0$ **i.e.** $(x - 1)(x + 1) \ge 0$

$\Rightarrow x \le -1$ **or** $x \ge 1$

Also, $x^2 \le 4 \Rightarrow x^2 - 4 \le 0$ **i.e.** $(x - 2)(x + 2) \le 0$

$\Rightarrow -2 \le x \le 2$

From (2) and (3), we get the domain of $f = \{(-\infty, -1] \cup [1, \infty)\} \cap [-2, 2]$

$= [-2, -1] \cup [1, 2]$

Q.4 Answer:

$\Sigma \cot x \cot y = 1$

Solution:

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$x = \tan^{-1}\sqrt{\frac{a}{bc}(a+b+c)}$

$\Rightarrow \tan x = \sqrt{\frac{a}{bc}(a+b+c)} \quad ...(1)$

$y = \tan^{-1}\sqrt{\frac{b}{ac}(a+b+c)}$

$\Rightarrow \quad \tan y = \sqrt{\frac{b}{ac}(a+b+c)}$ **...(2)**

and $z = \tan^{-1}\sqrt{\frac{c}{ab}(a+b+c)}$

$\Rightarrow \quad \tan z = \sqrt{\frac{c}{ab}(a+b+c)}$ **...(3)**

Adding eqn (1), (2) & (3) we get

$\tan x + \tan y + \tan z = \frac{(a+b+c)^{3/2}}{\sqrt{abc}}$

$\Rightarrow \tan x + \tan y + \tan z = \tan x \cdot \tan y \cdot \tan z.$

or $\frac{1}{\tan y \cdot \tan z} + \frac{1}{\tan x \cdot \tan z} + \frac{1}{\tan x \cdot \tan y} = 1$

$\Rightarrow \cot y \cdot \cot z + \cot x \cdot \cot z + \cot x \cdot \cot y = 1$

$\Rightarrow \Sigma \cot x \cot y = 1$

Q.5 Answer:

$\sqrt{\frac{(x^2+1)}{(x^2+2)}}$

Solution:

Let $\alpha = \tan^{-1} \mathrm{x} \Rightarrow \tan \alpha = \mathrm{x}$

then $\cos \alpha = \frac{1}{\sqrt{1+\tan^2 \alpha}} = \frac{1}{\sqrt{1+x^2}}$

$\Rightarrow \cos\left(\tan^{-1} x\right)\} = \left\{\frac{1}{\sqrt{1+x^2}}\right\}$

So, $\cot^{-1} \cos\left(\tan^{-1} x\right) = \cot^{-1}\left\{\frac{1}{\sqrt{1+x^2}}\right\}$

Let $\cot^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \beta$

$\Rightarrow \cot \beta = \frac{1}{\sqrt{1+\mathrm{x}^2}}$

and $\sin \beta = \frac{1}{\sqrt{1+\cot^2 \beta}}$

$= \frac{\sqrt{1+x^2}}{\sqrt{x^2+1+1}} = \sqrt{\frac{x^2+1}{x^2+2}}$

$\Rightarrow \sin\left[\cot^{-1}\left\{\cos\left(\tan^{-1}\right)\right\}\right] = \sqrt{\frac{x^2+1}{x^2+2}}$

Q.6 Answer:

$\{-1, 1, 0\}$

Solution:

$\sin^{-1} x = 2\tan^{-1} x$

$\Rightarrow \sin^{-1} x = \sin^{-1}\left\{2x/\left(1+x^2\right)\right\}$

$\Rightarrow 2x/\left(1+x^2\right) = x$

$\Rightarrow x(x+1)(x-1) = 0 \Rightarrow x = \{-1, 1, 0\}$

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Q.7 Answer:

$2n$

Solution:

Since $0 \le \cos^{-1} x_i \le \pi, \therefore \cos^{-1} x_i = 0$ **for all i.**

$\therefore x_i = 1$ **for all** $i \quad \therefore \sum_{i=1}^{2n} x_i = 2n$

Q.8 Answer:

$\left(0, \frac{\pi^2}{4}\right]$

Solution:

$$y = \left(\cot^{-1} x\right)\left(\cot^{-1}(-x)\right)$$
$$= \cot^{-1}(x)\left(\pi - \cot^{-1}(x)\right)$$

Now $\cot^{-1}(x)$ and $\left(\pi - \cot^{-1}(x)\right) > 0$

Using A.M. $\ge$ **G.M., we get** $\frac{\cot^{-1} x + \left(\pi - \cot^{-1} x\right)}{2} \ge \sqrt{\left(\cot^{-1} x\right)\left(\pi - \cot^{-1} x\right)}$

$\Rightarrow 0 < \cot^{-1}(x)\left(\pi - \cot^{-1}(x)\right)$

$\le \left(\frac{\cot^{-1} x + \left(\pi - \cot^{-1} x\right)}{2}\right) = \frac{\pi^2}{4}$

$\Rightarrow 0 < y \le \frac{\pi^2}{4}$

Q.9 Answer:

$\tan^{-1}(\sqrt{n})$

Solution:

$\sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r(r+1)}}\right) = \tan^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{1+\sqrt{r}\sqrt{(r-1)}}\right)$

$= \tan^{-1}\sqrt{r} - \tan^{-1}(\sqrt{r-1})$

$\Rightarrow \sum_{r=1}^{n} \sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r}(r+1)}\right)$

$= \sum_{r=1}^{n}\left(\tan^{-1}\sqrt{r} - \tan^{-1}\sqrt{r-1}\right) = \tan^{-1}\sqrt{n}$

Q.10 Answer:

$\frac{1}{2}$

Solution:

Given that $\cos^{-1} x = 2\sin^{-1} x$

$\Rightarrow \quad \frac{\pi}{2} - \sin^{-1} x = 2\sin^{-1} x$

$\Rightarrow \quad \frac{\pi}{2} = 3\sin^{-1} x \Rightarrow \sin^{-1} x = \frac{\pi}{6}$

So, $x = \sin\frac{\pi}{6} = \frac{1}{2}$

Q.11 Answer:

$[\tan\sin\cos 1, \tan\sin\cos\sin 1]$

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Solution:

We have, $1 \leq \sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x \leq \frac{\pi}{2}$

$\Rightarrow \quad \sin 1 \leq \cos^{-1}\sin^{-1}\tan^{-1}x \leq 1$

$\Rightarrow \cos\sin 1 \geq \sin^{-1}\tan^{-1}x \geq \cos 1$

$\Rightarrow \quad \sin\cos\sin 1 \geq \tan^{-1}x \geq \sin\cos 1$

$\Rightarrow \tan\sin\cos\sin 1 \geq x \geq \tan\sin\cos 1$

$\therefore \quad x \in [\tan\sin\cos 1, \tan\sin\cos\sin 1]$

Q.12 Answer:

1

Solution:

Let, $\sin^{-1}x = \tan^{-1}y = \theta$

$\Rightarrow \quad x = \sin\theta$ and $y = \tan\theta$

$\frac{1}{x^2} = \frac{1}{\sin^2\theta} = \operatorname{cosec}^2\theta$

and $\frac{1}{y^2} = \frac{1}{\tan^2\theta} = \cot^2\theta$

$\Rightarrow \frac{1}{x^2} - \frac{1}{y^2} = \operatorname{cosec}^2\theta - \cot^2\theta = 1$

Q.13 Answer:

$\frac{2ac}{a^2-c^2}$

Solution:

Let $\tan^{-1}x = \alpha$ and

$\tan^{-1}y = \beta \Rightarrow \tan\alpha = x, \tan\beta = y$

The given system of equations is $a\tan\alpha + b\sec\alpha = c$ and $a\tan\beta + b\sec\beta = c$

$\therefore \alpha$ and β are the roots of $a\tan\theta + b\sec\theta = c$

$\Rightarrow (b\sec\theta)^2 = (c - a\tan\theta)^2$

$\Rightarrow \left(a^2 - b^2\right)\tan^2\theta - 2ac\tan\theta + c^2 - b^2 = 0$

$\Rightarrow \tan\alpha + \tan\beta = \frac{2ac}{a^2-b^2}$ and

$\tan\alpha\tan\beta = \frac{c^2-b^2}{a^2-b^2}$

$\Rightarrow x + y = \frac{2ac}{a^2-b^2}$ and $xy = \frac{c^2-b^2}{a^2-b^2}$

$\Rightarrow 1 - xy = \frac{a^2-c^2}{a^2-b^2} \Rightarrow \frac{x+y}{1-xy} = \frac{2ac}{a^2-c^2}$

Q.14 Answer:

$\frac{x+y}{1-xy}$

Solution:

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The given function is $\tan\frac{1}{2}\left[\sin^{-1}\frac{2x}{1+x^2}+\cos^{-1}\frac{1-y^2}{1+y^2}\right]$

$\therefore \tan\frac{1}{2}\left[\sin^{-1}\frac{2x}{1+x^2}+\cos^{-1}\frac{1-y^2}{1+y^2}\right]$

$=\tan\frac{1}{2}\left[2\tan^{-1}x+2\tan^{-1}y\right] \quad \left[\text{as } 2\tan^{-1}x=\sin^{-1}\frac{2x}{1+x^2}=\cos^{-1}\frac{1-x^2}{1+x^2}\right]$

$=\tan\frac{1}{2}\left[2\left(\tan^{-1}x+\tan^{-1}y\right)\right]=\tan\left[\tan^{-1}x+\tan^{-1}y\right]$

$=\tan\left[\tan^{-1}\frac{x+y}{1-xy}\right]=\frac{x+y}{1-xy}$

Q.15 Answer:

$x=1,y=2;x=2,y=7$

Solution:

Converting cos and sin into tan, we get, $\tan^{-1}x+\tan^{-1}\left(\frac{1}{y}\right)=\tan^{-1}\left(\frac{3}{1}\right)$

$\Rightarrow \tan^{-1}\left[\frac{x+(1/y)}{1-x(1/y)}\right]=\tan^{-1}3$

$\Rightarrow \tan^{-1}\left(\frac{xy+1}{y-x}\right)=\tan^{-1}3 \Rightarrow \frac{xy+1}{y-x}=3$

$\Rightarrow x=\frac{3y-1}{y+3}=3-\frac{10}{y+3}$

So, for $x=1,y=2$ **and for** $x=2,y=7$

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